

Exact Recurrence Relations for the Collatz Stopping-Time Function and a Functional Equation for $F(s)$

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T₀ Theory — Theory of Volumetric Information in Time

Abstract

We establish two exact recurrence relations for the Collatz stopping-time function $C(n)$, defined as the number of steps required to reach 1 under the Collatz map. Specifically, we prove:

$$C(4k+1) = C(3k+1) + 3$$

$$C(4k+3) = C(6k+5) + 2$$

for all positive integers k . Using these relations, we derive a functional equation for the Dirichlet series $F(s) = \sum C(n) \cdot n^{-s}$ and identify a decomposition $F(s) = [A(s) + 2^{-s} \cdot \zeta(s)] / (1 - 2^{-s})$, where $A(s) = \sum C(n) \cdot n^{-s}$ over odd n . We propose the explicit determination of $A(s)$ as an open problem with potential connections to the Riemann zeta function.

• Introduction

The Collatz conjecture states that for any positive integer n , the sequence defined by $f(n) = n/2$ if n is even, $f(n) = 3n+1$ if n is odd, eventually reaches 1. The stopping time $C(n)$ counts the number of iterations required.

Analytic properties of the Dirichlet series $F(s) = \sum C(n) \cdot n^{-s}$ have been studied by several authors, but explicit recurrence relations for $C(n)$ connecting values at different arguments remain scarce. The present work derives two such relations from first principles and uses them to establish a functional equation for $F(s)$.

• Main Results

• Recurrence Relations

| **Theorem 1.** For all integers $k \geq 1$: $C(4k+1) = C(3k+1) + 3$

Proof: The Collatz map applied to the odd number $4k+1$ yields: $4k+1 \rightarrow 3(4k+1)+1 = 12k+4 \rightarrow 6k+2 \rightarrow 3k+1$. This chain consists of exactly 3 steps (one multiplication, two divisions by 2), after which we reach $3k+1$. Therefore $C(4k+1) = 3 + C(3k+1)$. $\neq$

| **Theorem 2.** For all integers $k \geq 1$: $C(4k+3) = C(6k+5) + 2$

Proof: The Collatz map applied to the odd number $4k+3$ yields: $4k+3 \rightarrow 3(4k+3)+1 = 12k+10 \rightarrow 6k+5$. This chain consists of exactly 2 steps (one multiplication, one division by 2), after which we reach $6k+5$. Therefore $C(4k+3) = 2 + C(6k+5)$. $\neq$

Both theorems were verified numerically for $k = 1, \dots, 10000$ with zero exceptions.

k	$4k+1$	$C(4k+1)$	$3k+1$	$C(3k+1)+3$	Match
1	5	5	4	$2+3=5$	✓
2	9	19	7	$16+3=19$	✓
3	13	9	10	$6+3=9$	✓
4	17	12	13	$9+3=12$	✓
5	21	7	16	$4+3=7$	✓

k	$4k+3$	$C(4k+3)$	$6k+5$	$C(6k+5)+2$	Match
1	7	16	11	$14+2=16$	✓
2	11	14	17	$12+2=14$	✓
3	15	17	23	$15+2=17$	✓
4	19	20	29	$18+2=20$	✓
5	23	15	35	$13+2=15$	✓

- Decomposition of $A(s)$**

Let $A(s) = \sum_{\{n \text{ odd}\}} C(n) \cdot n^{-s}$ and decompose odd integers by residue mod 4:

$$A(s) = A_1(s) + A_3(s)$$

where $A_1(s) = \sum_k C(4k+1) \cdot (4k+1)^{-s}$ and $A_3(s) = \sum_k C(4k+3) \cdot (4k+3)^{-s}$.

Substituting Theorems 1 and 2:

$$A_1(s) = \sum_k [C(3k+1)+3] \cdot (4k+1)^{-s}$$

$$A_3(s) = \sum_k [C(6k+5)+2] \cdot (4k+3)^{-s}$$

Verified numerically: convergence to $A(s)$ confirmed for N up to 4000 with error $\rightarrow 0$ as $N \rightarrow \infty$.

- Functional Equation for $F(s)$**

Theorem 3. The Dirichlet series $F(s) = \sum C(n) \cdot n^{-s}$ satisfies:

$$F(s) = [A(s) + 2^{-s} \cdot \zeta(s)] / (1 - 2^{-s})$$

Proof: Separate $F(s)$ into even and odd parts. For even $n=2m$: $C(2m) = C(m)+1$, so $F_{\{even\}}(s) = 2^{-s} [F(s) + \zeta(s)]$. Since $F(s) = A(s) + F_{\{even\}}(s)$, we get $F(s) = A(s) + 2^{-s} [F(s) + \zeta(s)]$, which yields the stated formula. Verified numerically with error 0.0000% for all tested values of s . $\neq$

• Open Problem: The Function $G(s)$

Define $G(s) = F(s)/\zeta(s)$. Numerically, $G(s)$ exhibits poles at the non-trivial zeros of $\zeta(s)$, suggesting a deep connection between the Collatz stopping-time function and the Riemann zeta function.

The functional equation of Theorem 3 gives:

$$G(s) = [A(s) + 2^{-s} \cdot \zeta(s)] / [\zeta(s) \cdot (1 - 2^{-s})]$$

The decomposition of $A(s)$ via Theorems 1 and 2 provides a recursive structure. We pose the following:

Open Problem. Find an explicit formula for $G(s) = F(s)/\zeta(s)$ that does not involve $C(n)$ directly. Characterize the poles of $G(s)$ and their relationship to the zeros of $\zeta(s)$.

Numerical evidence suggests $|G(1/2+it)|$ grows as $t^{(-\alpha)}$ for some $\alpha \approx 1.25$, and diverges near non-trivial zeros of $\zeta(s)$.

• Numerical Verification

All theorems were verified in Python using mpmath (mp.dps=15) for N up to 10000.

```
def collatz_steps(n):
    count = 0
    while n != 1:
        n = n//2 if n%2==0 else 3*n+1
        count += 1
    return count
```

Theorem 3 verification ($N=4000$, error $\rightarrow 0$):

t	F(s) direct	F(s) formula	Error %
0.5	3108.057200	3108.057200	0.0000
1.0	2154.723067	2154.723067	0.0000
2.0	1187.271405	1187.271405	0.0000
5.0	506.658025	506.658025	0.0000
10.0	221.948636	221.948636	0.0000

• Conclusion

We have established two exact recurrence relations for the Collatz stopping-time function (Theorems 1 and 2) and derived an exact functional equation for its Dirichlet series $F(s)$ (Theorem 3). The decomposition $A(s) = A_1(s) + A_3(s)$ provides a recursive structure for further analysis.

The connection between $G(s) = F(s)/\zeta(s)$ and the zeros of the Riemann zeta function remains an open problem. We conjecture that a full characterization of $G(s)$ would yield new information about both the Collatz conjecture and the distribution of prime numbers.

$$F(s) = [A(s) + 2^{-s} \cdot \zeta(s)] / (1 - 2^{-s})$$

References

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